

Course Announcements

Will add on Monday

Data Science Ethics

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ETHICS

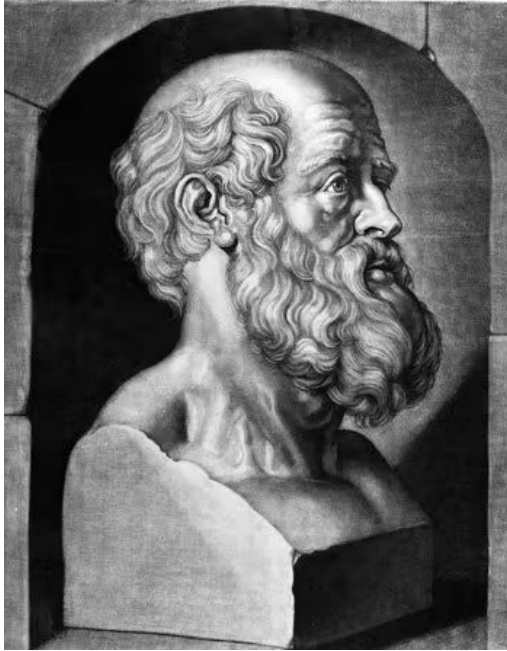
“Moral principles that govern a person's behaviour or the conducting of an activity.”

Ethical principles

Some modern principles that have stood up to lots of arguments:

- **Autonomy** (respect other's self determination)
- **Beneficence** (do good for others when possible)
- **Non-maleficance** (do not harm others)
- **Justice** (treat people equitably / equally / fairly)

Intent is not required for malfeasance. Harms / benefits / equity are subjective, not objective, and reasonable people will disagree about many things.



Professional ethics

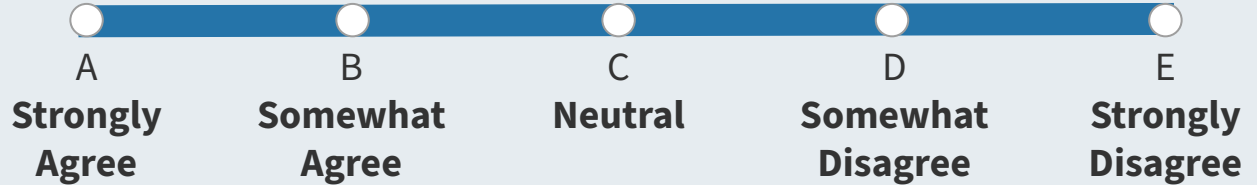
The Hippocratic Oath (Medicine)

- First do no harm
- Preserve patient privacy
- Do not be afraid to say I don't know
- Obligations to ALL human beings

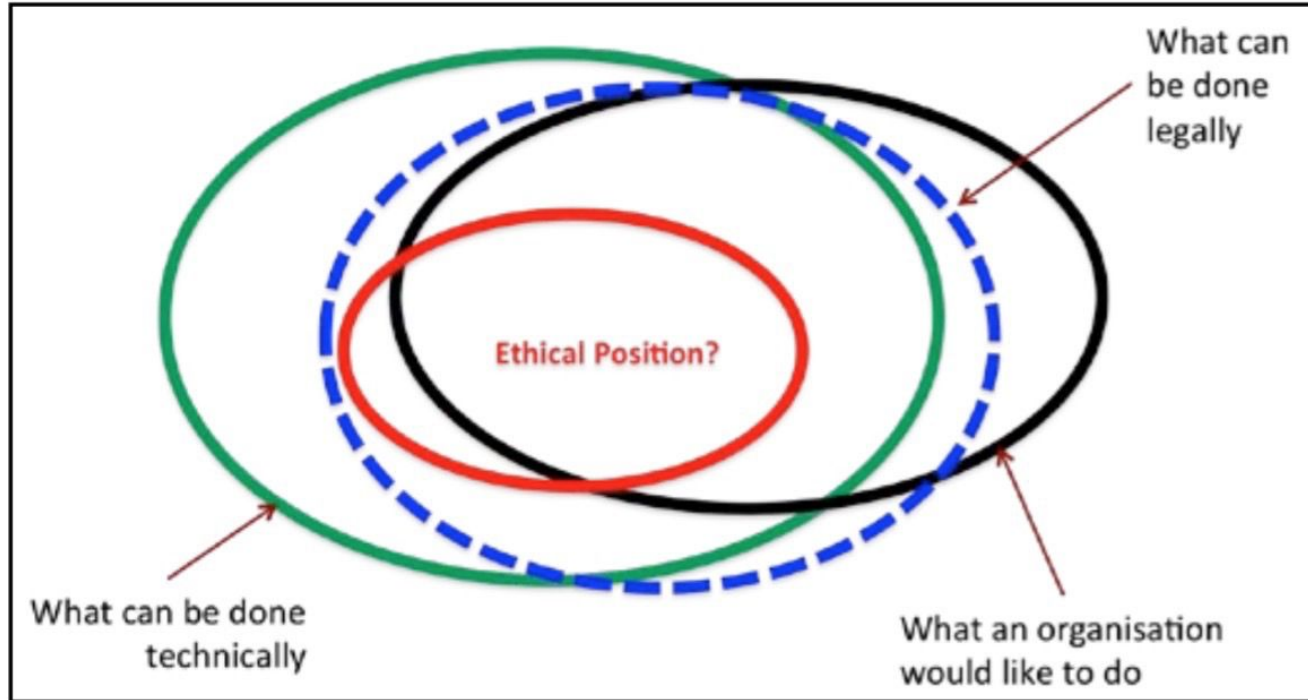
Data Science Ethics



When working on a data science project, data privacy is the main ethical concern.



Big Data Ethics



Ethical Data Science

Data science pursued in a manner so that is equitable, with respect for privacy and consent, so as to ensure that it does not cause undue harm.

Ethics is not a one-off lecture

Learn it early.

Return to it often.

Consider ethics at **every stage of the data science cycle**:

question → data collection → wrangling → analysis → communication (→ deployment)

YouTube vows to recommend fewer conspiracy theory videos

Site's move comes amid continuing pressure over i platform for misinformation and extremism

The Reason This "Racist Soap Dispenser" Doesn't Work on Black Skin

Amazon Prime and the racist algorithms

**MACHINES TAUGHT BY PHOTOS
LEARN A SEXIST VIEW OF
WOMEN**

Facial recognition software is biased towards white men, researcher finds

Biases are seeping into software

**YouTube's Restricted Mode Is Hiding
Some LGBT Content [Update]**

**Google Translate's Gender
Problem (And Bing Translate's,
And Systran's...)**

COGS 9 Examples

- Ashley Madison Hack [\[link\]](#)
- OKCupid Data Published [\[link\]](#)
- Equifax Hack [\[link\]](#)
- Google & Pentagon Team Up on Drones [\[link\]](#)
- Cambridge Analytica Data Breach To Influence US Elections [\[link\]](#)
- Amazon and Police Team Up on Facial Recognition & Surveillance [\[link\]](#)
- Amazon scraps secret AI recruiting tool biased against women [\[link\]](#)

A few additional examples we've compiled recently...

- Study of bias in AI [\[link\]](#)
- Pasco County Algorithmic Bias [\[link\]](#)
- Ethical issues (misogyny, racism) in large available dataset [\[link\]](#)
- Florida COVID-19 dashboard data scientist debacle [\[link\]](#)
- Banjo surveillance via fake apps [\[link\]](#)
- Google fires AI ethics founder [\[link\]](#) & Timnit Gebru's firing [\[link\]](#)
- MS lays off their entire ethics teams [\[link\]](#)
- ChatGPT is dumber than you think [\[link\]](#)
- Synthetic Media Creates New Social Engineering Threats [\[link\]](#)
- Generative AI is a copyright nightmare [\[link\]](#)
- GM allows drivers to be spied on [\[link\]](#)
- (Inaccurate) AI usage in official Deloitte report [\[link\]](#)
- Bartz v. Anthropic settlement on copyright [\[link\]](#)

1. THE QUESTION
2. THE DATA
3. ANALYSIS & EVALUATION
4. COMMUNICATION
5. AFTER “COMPLETION”

**CONSIDER
ETHICS
THROUGHOUT
THE ENTIRE
DATA SCIENCE
LIFE CYCLE**

1. THE QUESTION

- What is your question? Is it well-posed?
- Do you know something about the context and background of your question?
- What is the scope your investigation? What correlates might you inadvertently track? Is it possible to answer this question well?
- Who is asking you to answer this question? Are they open to all possible answers? What do they seek to gain?

Media file



Racial Photograph

[View media page](#)

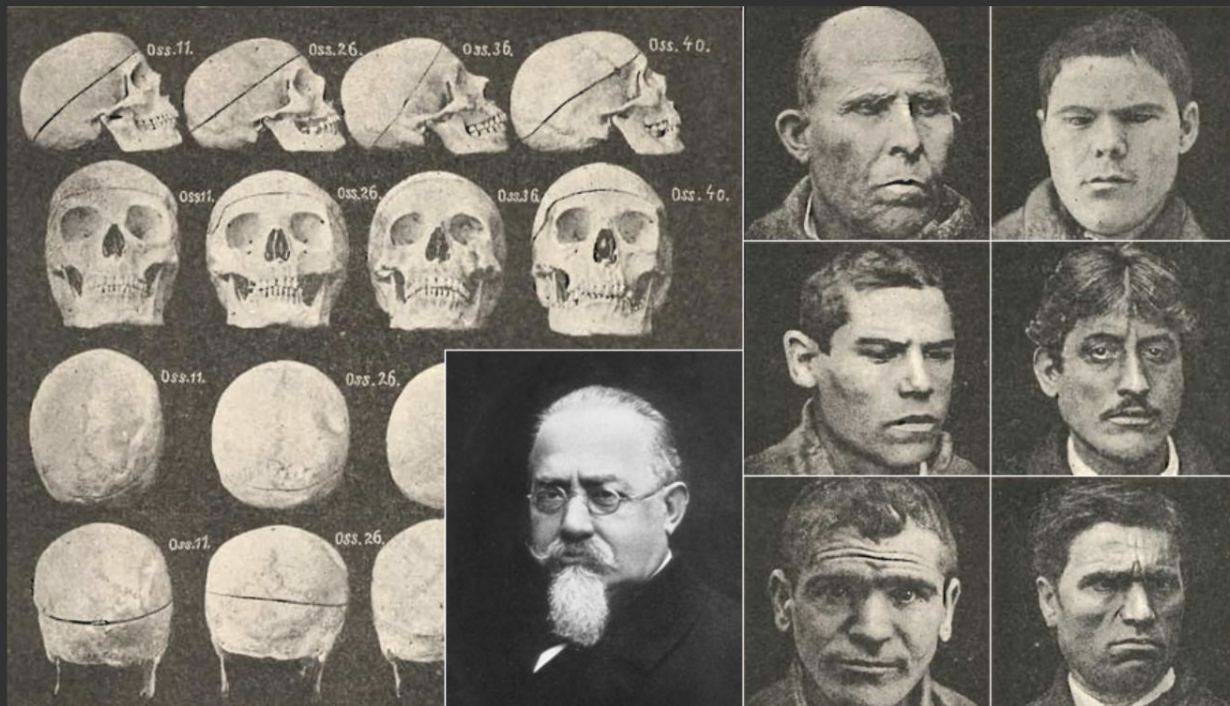
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Citations of this media

"Origin of Criminology"

From its inception photography had a profound effect on anthropological work as measuring device when studying races. While many would expect the photograph to bring truth and an end to the accentuated stereotyping by hand-drawn image, the medium would still be used to promote ideas of racial inferior traits. For example, Carl Victor and Friedrich Wilhelm Dammann's photographic book, *Races of Men* has influenced and propelled the viewpoints and stereotypes of different races. Containing black and white photos along with brief captions describing physical and mental traits, the context of these depictions serve to relay the idea of a Darwinian racial evolution from the Polynesians culminating with the Germanic race. Alphonse Bertillon founded modern anthropometric photography for the purpose of identifying repeated offenders by photographing and recording measures of physical features that remain constant throughout an individual's adult life. Cesare Lombroso, the founder of anthropological criminology, claimed to identify a links between common physical and mental traits and those highly likely to commit crimes. Dubbing the concept of being a "born criminal" Lombroso argued in favor of biological determinism. He found that skull and facial features were clues to genetic criminality and could be measured into quantitative research. The image depicts some of the 14 traits of a criminal Lombroso identified as large jaws, forward projection of jaw, low sloping forehead; high cheekbones, flattened or upturned nose; handle-shaped ears; hawk-like noses or fleshy lips; hard shifty eyes; scanty beard or baldness; insensitivity to pain; long arms, and so on. Lombroso viewed criminality as a hereditary disposition due to having traits similar to primitive human ancestors of monkeys and apes. His theories have also helped with influencing eugenics and anti-miscegenation laws, while his legacy can be found in modern day policing with racial profiling. "

—from "The Origins of Criminology"



Details

Scalar URL	https://scalar.usc.edu/works/measuring-prejudice/media/racial-photograph (version 1)
Source URL	https://scalar.usc.edu/works/measuring-prejudice/media/racial%20photography.jpg (image/JPEG)
dcterms:title	Racial Photograph
View as	RDF-XML, RDF-JSON, or HTML

This stuff just doesn't go away...

ARTICLE



<https://doi.org/10.1038/s41467-020-18566-7>

OPEN

Tracking historical changes in trustworthiness using machine learning analyses of facial cues in paintings

Lou Safra ^{1,2,3}✉, Coralie Chevallier¹, Julie Grèzes¹ & Nicolas Baumard ²✉

Received: 19 May 2019; Accepted: 10 August 2020;

Published online: 22 September 2020

Detecting criminality from faces [[link](#), [paper](#)]

Detecting Sexual Orientation From Faces with computer vision [[link](#), [paper](#)]

<https://www.nature.com/articles/s41467-020-18566-7.pdf>

2. THE DATA

Availability

- Are the data available? Who do you have data from?
- Do you have consent to use the data? Do you need consent?
- Do you have the data you need?
- What biases do your data have?

Privacy

- Can you guarantee privacy? Do you need to?
- What is the level of risk of your data, and how will you mitigate the risks? Are all subjects equally vulnerable?
- Are the data anonymized? Do they need to be?
- Are the data securely stored, with appropriate access rights

INFORMED CONSENT: the voluntary agreement to participate in research, in which the subject has an understanding of the research and its risks

Informed consent can be withdrawn at any point in time



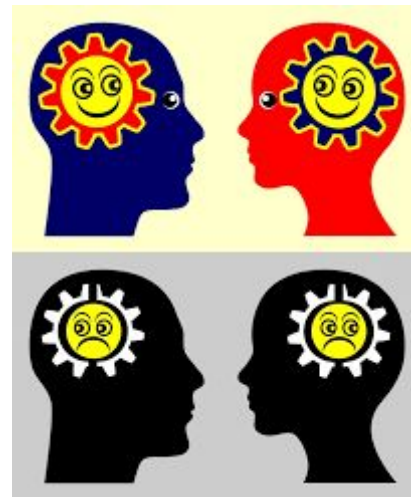
Think-Pair-Share

1. Is accepting terms of service on a social media platform equivalent to giving informed consent? Why or Why not?
2. If the data are available, are they OK to use? Consider concrete examples to support your conclusion.



Case Study: Emotional Contagion

Facebook conducted an experiment investigating whether they could manipulate people's emotions by manipulating the content displayed on one's newsfeed. [[link](#), [paper](#)]



Data Accuracy: Who's in the data?

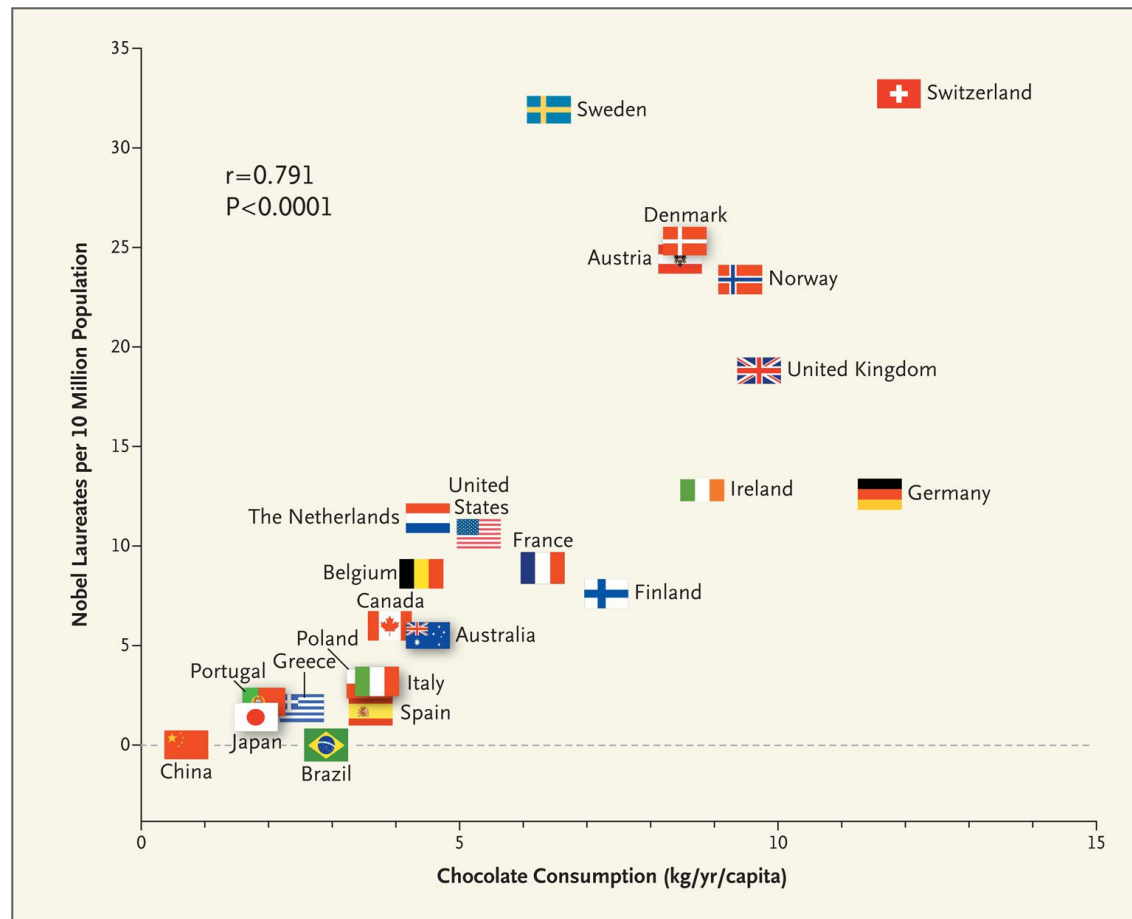
- Sampling bias (do you have everyone you need represented?)
- Historical bias
- Concerns with proxy measurements (If not directly measuring, limitations?)

CASE: Dissecting racial bias in an algorithm used to manage the health of populations

- Outcome Variable: Widely-used healthcare algorithm used **cost** as a proxy for **need**
- Historical bias: unequal access to care means that we spend less money caring for Black patients than for White patients
- Outcome: Systematically underestimated how sick Black patients were

3. ANALYSIS & EVALUATION

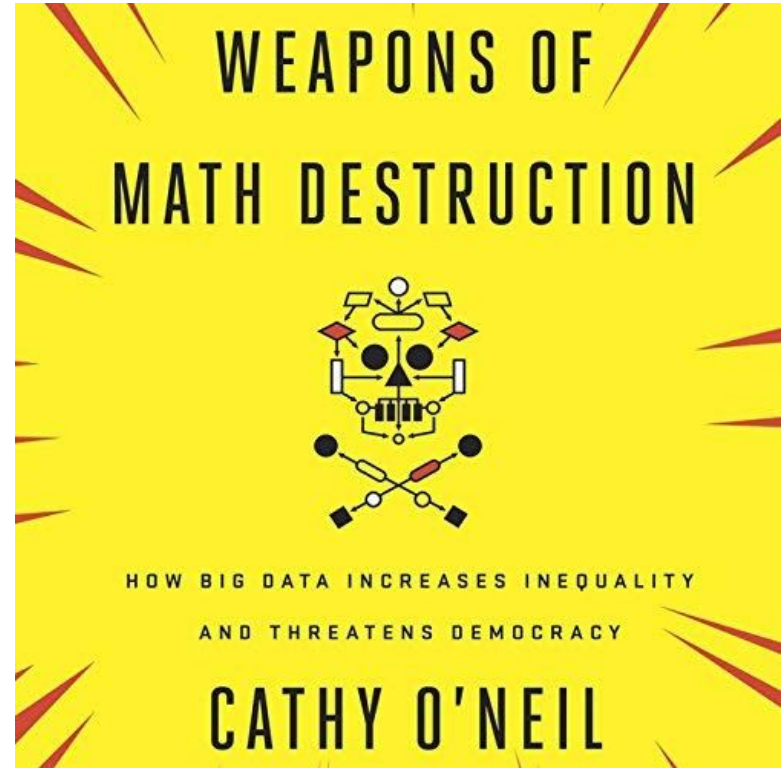
- Have you avoided common pitfalls? p-hacking , garden of forking paths, dropping inconvenient data without cause,
- Are the outcomes/metrics being used accurate measures?
 - a. Goodhart's Law: when a measure becomes a target, it ceases to be a good measure.
- Do your analyses reflect spurious correlations? Establish causality?
- What kind of covariates might need to be included?
- Is your analysis reproducible? Replicable?
 - a. **Reproducible**: Can you get the same answers that I did when you analyze my data?
 - b. **Replicable**: Can you get the same answers that I did when you collect your own data?



Case Study: Teacher Rating

Washington, DC school district used an algorithm to rate teachers, based on test score growth. Scores from this algorithm were used to fire ‘low performers’.

They had no independent measure of teaching quality.



On INTENT and OBJECTIVITY

- Intent is not required for harmful practices to occur
- Data, algorithms and analysis are not objective.
 - It is done by people, who have biases
 - It uses data, which have biases
- Data Science is powerful
- Bias & discrimination driven by data & algorithms can give new scale to pre-existing inequities

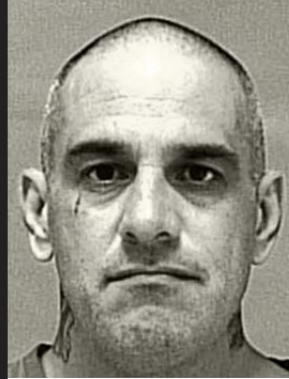
4. COMMUNICATION

- Have you effectively and accurately communicated your results? Are the necessary parties informed?
- Are limitations clearly identified?
- Is your model a black box? Is it interpretable as to how it came to any particular decision?
- Is there a way to appeal a model decision? What kind of evidence would you need to refute a decision?

Case Study: Predictive Policing

- Predictive policing uses algorithms to predict crime, and recidivism
- Input data can be highly correlated [\[link\]](#) with race & SES, reflecting spurious correlations and leading to discriminatory decisions.
- These algorithms and decisions are often opaque and un-appealable.

Two Petty Theft Arrests

	
VERNON PRATER	BRISHA BORDEN
RISK: 3	RISK: 8

Borden was rated high risk for future crime after she and a friend took a kid's bike and scooter that were sitting outside. She did not reoffend.

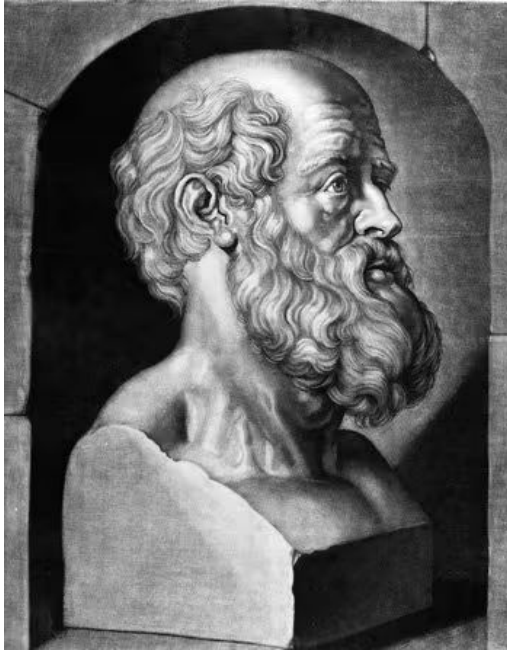
5. AFTER “COMPLETION”

Healthy models maintain a back and forth with the thing(s) in the world they are trying to understand.

- Are you proactively looking for potential unintended side effects of your model itself or harmful outputs?
- Do you have a mechanism to fix and update your algorithm?

ON PERPETUATING INEQUALITY

- Data & Algorithms can & will entrench social disparities
- Errors and bias typically target the disenfranchised
- The combination of damage, scale, and opacity can be incredibly destructive
- They can introduce feedback in such a way as to enact self-fulfilling prophecies



Professional ethics

Data Science Oath?

- First do no harm
- Preserve patient privacy
- Do not be afraid to say I don't know
- Obligations to ALL human beings

BOX D.1

Hippocratic Oath

I swear to fulfill, to the best of my ability and judgment, this covenant:

I will respect the hard-won scientific gains of those physicians in whose steps I walk, and gladly share such knowledge as is mine with those who are to follow.

I will apply, for the benefit of the sick, all measures which are required, avoiding those twin traps of overtreatment and therapeutic nihilism.

I will remember that there is art to medicine as well as science, and that warmth, sympathy, and understanding may outweigh the surgeon's knife or the chemist's drug.

I will not be ashamed to say "I know not," nor will I fail to call in my colleagues when the skills of another are needed for a patient's recovery.

I will respect the privacy of my patients, for their problems are not disclosed to me that the world may know. Most especially must I tread with care in matters of life and death. If it is given me to save a life, all thanks. But it may also be within my power to take a life; this awesome responsibility must be faced with great humbleness and awareness of my own frailty. Above all, I must not play at God.

I will remember that I do not treat a fever chart, a cancerous growth, but a sick human being, whose illness may affect the person's family and economic stability. My responsibility includes these related problems, if I am to care adequately for the sick.

I will prevent disease whenever I can, for prevention is preferable to cure.

I will remember that I remain a member of society, with special obligations to all my fellow human beings, those sound of mind and body as well as the infirm.

If I do not violate this oath, may I enjoy life and art, respected while I live and remembered with affection thereafter. May I always act so as to preserve the finest traditions of my calling and may I long experience the joy of healing those who seek my help.

SOURCE: L.C. Lasagna, 1964, *Hippocratic Oath*, Modern Version, The Johns Hopkins Sheridan Libraries and University Museums. <http://guides.library.jhu.edu/c.php?g=202502&p=1335759>, accessed August 21, 2017.

BOX D.2

Data Science Oath

I swear to fulfill, to the best of my ability and judgment, this covenant:

I will respect the hard-won scientific gains of those data scientists in whose steps I walk and gladly share such knowledge as is mine with those who follow.

I will apply, for the benefit of society, all measures which are required, avoiding misrepresentations of data and analysis results.

I will remember that there is art to data science as well as science and that consistency, candor, and compassion should outweigh the algorithm's precision or the interventionist's influence.

I will not be ashamed to say, "I know not," nor will I fail to call in my colleagues when the skills of another are needed for solving a problem.

I will respect the privacy of my data subjects, for their data are not disclosed to me that the world may know, so I will tread with care in matters of privacy and security. If it is given to me to do good with my analyses, all thanks. But it may also be within my power to do harm, and this responsibility must be faced with humbleness and awareness of my own limitations.

I will remember that my data are not just numbers without meaning or context, but represent real people and situations, and that my work may lead to unintended societal consequences, such as inequality, poverty, and disparities due to algorithmic bias. My responsibility must consider potential consequences of my extraction of meaning from data and ensure my analyses help make better decisions.

I will perform personalization where appropriate, but I will always look for a path to fair treatment and nondiscrimination.

I will remember that I remain a member of society, with special obligations to all my fellow human beings, those who need help and those who don't.

If I do not violate this oath, may I enjoy vitality and virtuosity, respected for my contributions and remembered for my leadership thereafter. May I always act to preserve the finest traditions of my calling and may I long experience the joy of helping those who can benefit from my work.

Activity: pre-mortem

Consider this (fictional) COGS 108 Project: Campus dining gave us a year of meal-swipe counts and food-waste logs for every dining location. We'll build a model of waste per swipe and recommend reducing hours or menu items wherever waste per swipe is highest. Less waste is better for the planet and saves money.

*It's October 2027. Your COGS 108 project is in the news for the wrong reasons. **Write the headline.***



Further resources...

For a book-length treatment of ethical issues in statistics, see **Hubert and Wainer** (2012). The **National Academies** report on data science for undergraduates National Academies of Science, Engineering, and Medicine (2018) included data ethics as a key component of data acumen. The report included the draft oath for data scientists.

A historical perspective on the ASA's Ethical Guidelines for Statistical Practice can be found in Ellenberg (1983). The University of Michigan provides an EdX course on "[Data Science Ethics](#)." **Carl Bergstrom** and **Jevin West** developed a course Calling Bullshit: Data Reasoning in a Digital World". Course materials and related resources can be found at <https://callingbullshit.org>.

Andrew Gelman has written a column on ethics in statistics in *CHANCE* for the past several years (see, for example Andrew Gelman (2011); Andrew Gelman and Loken (2012); Andrew Gelman (2012); Andrew Gelman (2020)). *Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy* describes a number of frightening misuses of big data and algorithms (O'Neil 2016).

The **Teach Data Science** blog has a series of entries focused on data ethics (<https://teachdatascience.com>). D'Ignazio and Klein (2020) provide a comprehensive introduction to data feminism (in contrast to data ethics). The ACM Conference on Fairness, Accountability, and Transparency (FACCT) provides a cross-disciplinary focus on data ethics issues (<https://facctconference.org/2020>).

Deon has made an [ethics checklist](#) for data science projects. Note that checklists are a good place to start...but not exhaustive nor an excuse to avoid critical thinking.

The **Center for Open Science**—which develops the [Open Science Framework](#) (OSF)—is an organization that promotes openness, integrity, and reproducibility in scientific research. The OSF provides an online platform for researchers to publish their scientific projects. **Emil Kirkegaard** used OSF to publish his OkCupid data set.

The **Institute for Quantitative Social Science** at Harvard and the **Berkeley Initiative for Transparency in the Social Sciences** are two other organizations working to promote reproducibility in social science research. The **American Political Association** has incorporated the [Data Access and Research Transparency](#) (DA-RT) principles into its ethics guide. The Consolidated Standards of Reporting Trials (CONSORT) statement at (<http://www.consort-statement.org>) provides detailed guidance on the analysis and reporting of clinical trials.

Many more examples of how irreproducibility has led to scientific errors are available at <http://retractionwatch.com/>. For example, [a study linking severe illness and divorce rates](#) was retracted due to a coding mistake.

Data Science Ethics (post)



When working on a data science project, data privacy is the main ethical concern.

